DS 4420: Game Player Retention Analysis

Sharyq Siddiqi, Nathan Moeykens, Leyuan Wang

Northeastern University

Introduction/Abstract

With the online game industry growing over the past few decades, a focal point in many game studios is what types of players will be drawn and stay with a given games player base. With modern games generating large amounts of gameplay data: machine learning and data analysis can be used to identify patterns in players behavior to understand what traits may be related to retaining a player. This project explores how gameplay metrics and player activity relate to player retention in online games through 3 different types of models(MLP, Bayesian, and Time Series). Due to the metric in which each model could predict these traits being slightly different, with each model having their own strengths, the MLP had a accuracy of 69.6%, the Collaborative Filtering had an accuracy of 72.5% and the Bayesian model has an estimated R^2 of 0.334, all being relative to the training data of each model.

Data Collection

In order to predict player engagement and churn, data that contains player and game statistics is needed in a time series format. The data also needs to be collected from a large and popular game with an API to support the collection, and so the Riot Games and their League of Legends API was eventually chosen to use. In order to track player retention, players and their match history were needed, so random samples of active players were taken from certain leaderboards. From here, each player's match statistics were collected and aggregated. Only values useful for our analysis were chosen such as player scores like kills and deaths, and overall game statistics like match length and winner. To create the time series data, match time was aggregated by day with values being 0 if no games were played that day. In general, these attributes allow for the data to be utilized with Collaborative Filtering, Bayesian, and MLP models, and will directly address player churn and engagement can predict if users will be active on a game and why.

Methodology

For the Bayesian Machine Learning Model, it was found that average time played in a given match was the closest pseudo-metric that could be used to determine a single player's engagement. With this data, a Log-Normal Likelihood Distribution was built using the BRMS R Librarys:

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} exp(-\frac{1}{2}(\frac{log(y) - \mu}{\sigma})^2)$$

This distribution captures two qualities essential for predicting time in this case: the result being non-negative like the exponential, and the playtime not being something that will always increase in magnitude like the gaussian. In the final model, 9 features from the data are considered(kills, deaths, assists, vision score, total heal, total heals on teammates, win ratio, early surrender ratio), with the inclusion of kills and deaths being a fixed feature point to monitor in conjunction with each other. Traits excluded were found to be collinear. Normal Distributions were used for all the priors. The model was trained on 4 chains, 2000 iterations, and an 18% burn(320 iterations), and a thinning value of 2. A Train/Test split was used for cross validating

The collaborative filtering model utilized time series data to predict how long users will play on a given day, if at all. Instead of just calculating similarity scores, users must be compared to others n days in the past, to predict n days in the future. The other main difference between this filtering model and a typical one is the time series data is not centered to preserve zero predictions

Methodology Cont.

The Multilayer Perceptron (MLP) model was used to predict player retention from recent gameplay behavior. Features were constructed using a rolling window over recent matches, including statistics such as KDA, win rate, match frequency, champion diversity, and losing streak. A small neural network with one hidden layer was trained using gradient descent with a sigmoid output layer. Model performance was evaluated using a train/test split by player and standard classification metrics including accuracy, precision, recall, and F1-score.

Results

It was found that there was heavily correlation with a players vision control within game, how often an early surrender was called and a negative correlation with kills relative to deaths. The other factors did also have some impact, but with less confidence associated with it. The RMSE found it was about 147.89, implying that the model was accurate within a 147 margin of error usually, which in the context of games that can extend 30 minutes is relatively impressive. The R^2 score found on the test data achieved showed an estimated value of 0.334, showing a lower level of variance. In the Figure are 4 random predictions by the model for 4 different players. In blue is the curve, and the red x shows the players actual value.

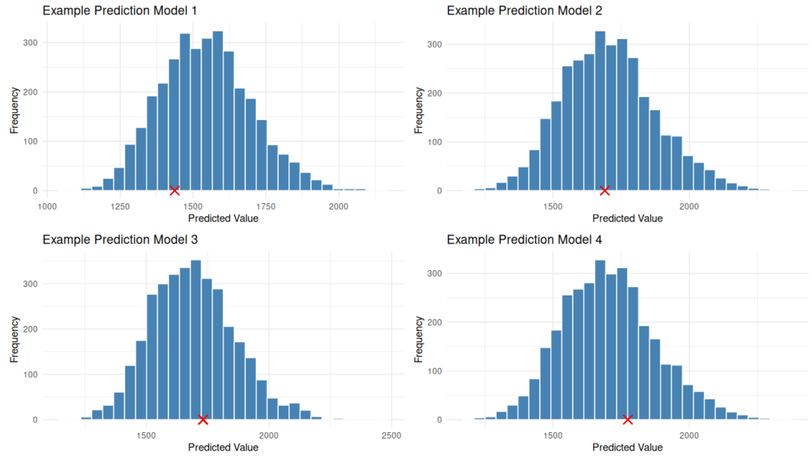
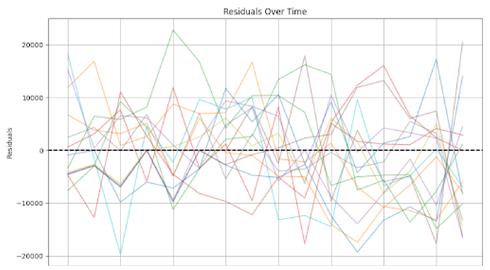


Fig. 1: Example Predictions for Log-Normal Bayesian Model

The collaborative filtering model correctly predicted whether a user would play or not on a given day with 72.5% accuracy. This score held steady from predicting one day to five days. In terms of predicting the length a user would play on a certain day, the R^2 was negative, meaning the prediction was worse than simply guessing the average. This is mainly due to the sparsity of the data and lack of collected games that gave misleading time series entries. As in the residual plot, the predictions were neither overestimated or underestimated in an obvious way.



Results Cont.

An MLP model was trained to predict player retention using gameplay behavior features derived from recent match history. The best-performing configuration used 8 hidden nodes, a learning rate of 0.01, and 6000 training iterations. Test performance: Accuracy 69.6%, Precision 95.3%, Recall 71.3%, F1 Score 81.6%.

Due to strong class imbalance in the dataset (94% retained players), a simple majority baseline achieves higher overall metrics. Nevertheless, the MLP converges during training and captures some behavioral signal from gameplay statistics.

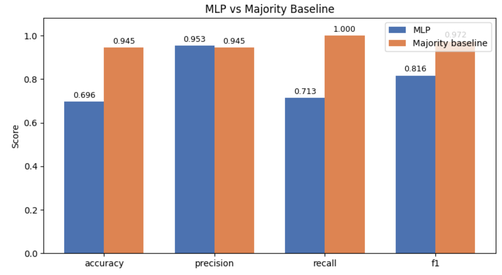


Fig. 3: MLP vs Majority Baseline

Conclusion/Future Work

This study investigated whether gameplay behavior can be used to understand and predict player retention in online games. Using match statistics collected from the Riot Games API, multiple behavioral features such as KDA, win rate, match frequency, and other gameplay indicators were analyzed.

Three modeling approaches were explored. The collaborative filtering time-series model achieved an accuracy of approximately 72.5% when predicting whether a player would return to play on a given day.

The Bayesian model was able to estimate player engagement through match playtime with an RMSE of about 147.89 and an R^2 of approximately 0.334.

The MLP model converged during training and achieved an accuracy of 69.6%, demonstrating that gameplay features contain some predictive signal related to player engagement.

However, the dataset exhibited strong class imbalance, with the majority of samples representing retained players. This imbalance, along with the sparsity of time-series gameplay data, limits the predictive power of the models.

Future work could improve the analysis by collecting larger datasets, incorporating richer behavioral and temporal features, and applying techniques designed to handle imbalanced classification problems. Combining behavioral statistics with more detailed temporal activity patterns may also improve predictions of long-term player retention.

References

[1] Paul Bürkner. *Parameterization of Response Distributions in BRMS*. Sept. 2025. URL: https://cran.r-project.org/web/packages/brms/vignettes/brms_families.html.